MIL-D-3994C
23 July 1982
SUPERSEDING
MIL-D-3994B
31 August 1970

MILITARY SPECIFICATION

DEXTRIN, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers technical grade dextrin.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-S-48 - Sacks, Shipping, Paper

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

FSC 6810

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D1193 - Reagent Water
D1921 - Particle Size (Sieve Analysis) of Plastic Materials

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Chemical and physical characteristics. Dextrin shall conform to the chemical and physical characteristics of table I when tested as specified therein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the dextrin produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

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TABLE I. Chemical and physical characteristics

Characteristic	Maximum percent by weight	Test paragraph
Particle size: material retained on a No. 80 sieve	0.5	4.2.4.1
Uncombined water	5.0	4.2.4.2
Ash	0.15	4.2.4.3
Water-insoluble matter	2.5	4.2.4.4
Acidity (as acetic acid)	0.18	4.2.4.5
Reducing sugars (as dextrose)	4.0	4.2.4.6

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For dextrin test. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 1/2 pound shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for dextrin test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.2.3 For container leakage test. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

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(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible
(f) Unitization not as specified

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 1 pound. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.3.3 For container leakage test. The sample unit shall be one container. The sample containers selected in 4.2.2.3 shall be tested as specified in 4.2.5 using an AQL of 1.5 percent defective.

4.2.4 Tests. Water in accordance with ASTM D1193, type as applicable, and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Particle size. Determine the particle size of the specimen in accordance with ASTM D1921, using a tared No. 80 sieve.

4.2.4.2 Uncombined water. Weigh to the nearest 0.01 gram (g) approximately 12 g of the specimen into a suitable tared dish and dry at 105° ± 3°C for 4 hours. Cool to room temperature in a desiccator and reweigh. Calculate the uncombined water as follows:

$$\text{Percent uncombined water} = \frac{100\ (A - B)}{W}$$

where: A = Weight in grams of the dish and contents before drying,
B = Weight in grams of the dish and contents after drying, and
W = Weight in grams of specimen.

Retain the residue for use in the water-insoluble matter test in 4.2.4.4.

4.2.4.3 Ash. Weigh to the nearest milligram approximately 5 g of the specimen into a tared silica or platinum crucible. Ignite at 600° ± 10°C until the ash is white. Cool to room temperature in a desiccator and reweigh. Calculate the percent ash as follows:

$$\text{Percent ash} = \frac{100\ (A - B)}{W}$$

where: A = Weight in grams of the crucible and ash,
B = Tare weight in grams of the crucible, and
W = Weight in grams of the specimen.

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4.2.4.4 Water-insoluble matter. Quantitatively wash the residue retained from 4.2.4.2 into a 250-milliliter (ml) beaker using 100 ml of hot (60° ± 5°C) water. Stir until complete solution is effected, filter through a tared filter crucible using suction, and wash with 90 ml of hot water. Heat the residue in the crucible at 105° ± 3°C for 4 hours. Cool and weigh. Calculate the percent water-insoluble matter as follows:

$$\text{Percent water-insoluble matter} = \frac{100\ (A - B)}{W}$$

where: A = Weight in grams of crucible and residue,
B = Tare weight in grams of crucible, and
W = Weight in grams of the specimen taken in 4.2.4.2 before drying.

4.2.4.5 Acidity (as acetic acid). Weigh into a flask to the nearest milligram approximately 6 g of the specimen. Add 100 ml of water and stir until solution is effected. (Ignore small quantities of insoluble matter that may be present.) Add a few drops of phenolphthalein indicator and titrate with 0.1N sodium hydroxide solution until the first pink color that persists for 30 seconds is produced. Perform a blank determination on 100 ml of the same water used to prepare the specimen solution. Calculate the acidity as follows:

$$\text{Percent acidity} = \frac{6N\ (A - B)}{W}$$

where: N = Normality of the sodium hydroxide solution,
A = Milliliters of sodium hydroxide solution used to titrate the specimen,
B = Milliliters of sodium hydroxide solution used to titrate the blank, and
W = Weight in grams of specimen.

4.2.4.6 Reducing sugars (as dextrose).

(a) Cupric sulfate solution. Dissolve 34.64 g of crystalline cupric sulfate pentahydrate in water to make 500 ml.

(b) Alkaline tartrate solution. Dissolve 173 g of potassium sodium tartrate tetrahydrate and 50 g of sodium hydroxide in water to make 500 ml.

(c) Procedure. Weigh 2.00 g of the specimen into a beaker. Add approximately 75 ml of water and shake for 10 minutes. Filter until the filtrate is clear. Dilute the filtrate with water to make 100 ml of solution. Mix equal volumes of the cupric sulfate solution prepared as specified in (a) and the alkaline tartrate solution prepared as specified in (b). Add 50 ml of the resulting solution to 50 ml of the specimen filtrate and boil for 3 minutes. Filter through a tared filter crucible. Wash the precipitate, first with water, then with alcohol, and finally with ether. Dry at 105° ± 3°C to

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constant weight. The weight of the precipitate shall be no greater than 0.0925 g. (This is equivalent to 4 percent by weight of dextrose in the 2.00 g specimen.)

4.2.5 Container leakage test. Place the container in each of the following positions, and leave it in each for a period of 15 minutes:

(a) Upright
(b) Upside down
(c) On one side
(d) On other side

Examine the container after each period for any evidence of leakage.

5. PACKAGING

5.1 Unit packing. Dextrin shall be unit packed level B or industrial as specified (see 6.2).

5.1.1 Level B. A quantity of 50 (+0.50 or -0) pounds (lb) of dextrin shall be unit packed in a sack conforming to type III or IV style B, sack construction number 5 or 5X MB grade 2 of UU-S-48. The sack shall be closed in accordance with the applicable requirements of UU-S-48. There shall be no evidence of leakage of dextrin from the sack when tested as specified in 4.2.5.

5.1.2 Industrial. A quantity of 50 (+0.50 or -0) lb of dextrin shall be unit packed industrially in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level B or industrial as specified (see 6.2).

5.2.1 Level B. Dextrin, unit packed level B as specified in 5.1.1, shall require no further protection for shipment other than unitization.

5.2.2 Industrial. Dextrin shall be packed industrially in accordance with MIL-STD-1188.

5.2.3 Unitization. A uniform quantity of level B packs of dextrin shall be palletized in accordance with the applicable requirements for load type XV, using a fiberboard cap, bonding by shrink-wrap, stretch wrap, or combined bonding means K and L of MIL-STD-147. Industrial packs shall be unitized in a manner to assure protection of packs and contents during transport from the supply source to the first destination, short term storage not exceeding 6 months, and acceptance by common carrier.

5.3 Marking. Level B containers shall be marked in accordance with MIL-STD-129. Industrial unit packs and packs shall be marked in accordance with MIL-STD-1188. Each container shall be marked to show lot or batch number and date of manufacture of the dextrin.

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6. NOTES

6.1 Intended use. Dextrin is intended for use in pyrotechnic mixtures and primary explosive mixtures.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of unit packing and packing required (see 5.1 and 5.2).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

Custodian:

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Preparing activity:

Army - EA

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